Two routes to the same future currency

Interest rate parity connects today's spot rate, interest rates in two currencies, and the forward exchange rate. If two fully hedged strategies deliver the same currency on the same date, their costs should agree. Otherwise, borrowing in one currency and investing in the other could create a covered arbitrage.

Use EURUSD, so both spot and forward rates are USD per EUR. Let S be today's spot rate, F the forward rate for maturity T, and G the growth factor over that period. Starting with EUR 1 gives two possible dollar amounts at maturity.

Route	Dollars at maturity
Convert now; invest in USD	S x G_USD
Invest in EUR; sell future EUR forward	G_EUR x F

$$F = S\frac{G_{\text{USD}}}{G_{\text{EUR}}}$$

The growth factors must use the same horizon and the rate convention actually quoted. This lecture's numerical examples use annual simple rates, so the growth factor is 1 + rT. For six months, T = 0.5. The later bond examples use semiannual compounding instead.

$$F = S\frac{1 + r_{\text{USD}}T}{1 + r_{\text{EUR}}T}$$

With EURUSD = 1.0850, a one-year USD rate of 5% and EUR rate of 3%, the fair one-year forward is 1.106068 USD per EUR. The higher USD interest rate is offset by the higher forward dollar price of the euro.

Forward premium is quote-specific
When EURUSD forward exceeds EURUSD spot, the euro is at a forward premium and the dollar is at a forward discount. This describes today's prices for different settlement dates; it does not prove that the euro will appreciate.

Lock a foreign-currency payment

A U.S. company must pay EUR 2.5 million in 18 months. If EURUSD rises, it will need more dollars to make the same euro payment. A long euro forward fixes that dollar cost today.

Input	Value
Payment	EUR 2,500,000
Time to payment	1.5 years
Spot EURUSD	1.0850 USD per EUR
USD annual simple rate	5.25%
EUR annual simple rate	3.75%

$$F = 1.0850\frac{1 + 0.0525(1.5)}{1 + 0.0375(1.5)} = 1.108112$$

The forward fixes a dollar payment of USD 2,770,281.07, calculated using the unrounded rate. The future spot rate determines whether the forward looks favorable after the fact, but it no longer changes the contracted dollar payment.

A money-market hedge reaches the same result

Alternatively, buy enough euros today that a euro investment grows to EUR 2.5 million. This requires EUR 2,366,863.91. Buy those euros at spot and finance their dollar cost until the payment date. The dollar financing grows at the USD rate.

$$\text{USD cost}_T = \frac{\text{EUR payment}_T}{1 + r_{\text{EUR}}T}\, S(1 + r_{\text{USD}}T)$$

Substituting the parity forward into the direct forward hedge produces exactly the same dollar amount. That equivalence is why the pricing formula works; it is not an arbitrary adjustment to spot. The forward bundles the economics of the currency conversion and the two money-market positions.

Choose the position from the cash flow
A euro payable requires buying euros forward. A euro receivable requires selling euros forward. The currency, amount and payment date determine the hedge.

Covered interest arbitrage

Consider GBPUSD = 1.25, a one-year USD rate of 5%, and a GBP rate of 2%. Parity implies a forward of 1.25 x 1.05 / 1.02 = 1.286765. Suppose you can instead sell pounds forward at 1.30. Future pounds are expensive relative to the covered investment route.

Transaction	Amount
Borrow USD today	USD 1,000,000
Buy GBP at spot	GBP 800,000
Invest GBP for one year at 2%	GBP 816,000 at maturity
Sell all future GBP forward at 1.30	USD 1,060,800
Repay USD borrowing at 5%	USD 1,050,000
Covered surplus	USD 10,800

Every exchange rate needed for repayment is fixed today. This is why the trade is covered. The opposite deviation reverses the currencies: if the base currency is too cheap forward, borrow it, sell it at spot, invest the quote currency, and buy the full base-currency repayment forward.

A higher foreign rate alone is not an arbitrage

Investing abroad without a forward leaves the final home-currency value exposed to the future spot rate. Depreciation can offset the higher interest return. Once the FX risk is covered, the forward adjustment normally absorbs the interest differential.

Why textbook parity and market quotes may differ

Actual borrowing and lending rates differ. Bid-ask spreads, collateral, credit limits and balance-sheet costs also matter. A gap calculated with frictionless mid-market inputs may not be an executable profit. Cross-currency basis can reflect funding and hedging conditions.

> **Check both sides at maturity**
> Include all investment proceeds and the full loan repayment, including interest. Cover the repayment amount, not just the original principal. Compare cash flows on the same date.

Further reading: BIS Quarterly Review, September 2016, "Covered interest parity lost: understanding the cross-currency basis."

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